

gemstone-rs Feature Map

Crates, examples, docs, and gemstone-py parity references



Generated from the Markdown sources in `gemstone-rs/docs`.

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gemstone-rs Feature Map

This map is the Rust equivalent of `gemstone-examples plan3-map` in `gemstone-py`: it ties each feature stream to the crates, examples, docs, and Python reference point that inspired it.

Run it from the CLI:

```
gemstone-rs hello
gemstone-rs compare gemstone-py
gemstone-rs compare gemstone-py --gaps
gemstone-rs compare gemstone-py --next
gemstone-rs compare gemstone-py --totals
gemstone-rs compare gemstone-py --batches
gemstone-rs compare gemstone-js
gemstone-rs compare gemstone-js --gaps
gemstone-rs compare gemstone-js --next
gemstone-rs compare gemstone-js --totals
gemstone-rs compare gemstone-js --batches
gemstone-rs compare all --next
gemstone-rs compare all --totals
gemstone-rs compare all --batches
gemstone-rs examples map
gemstone-rs examples map --json
gemstone-rs examples scaffold quickstart ./gemstone-rs-quickstart
gemstone-rs examples scaffold codegen_workflow ./gemstone-rs-codegen-workflow
gemstone-rs examples scaffold profile_codegen_workflow ./gemstone-rs-profile-codegen
gemstone-rs examples scaffold generated_wrapper_app ./gemstone-rs-generated-wrapper
gemstone-rs examples scaffold session_worker_pool ./gemstone-rs-worker-pool
gemstone-rs examples scaffold axum_service ./gemstone-rs-axum-service
gemstone-rs examples scaffold actix_service ./gemstone-rs-actix-service
```

`hello` is the no-live sanity check that mirrors `gemstone-examples hello`. `compare gemstone-py` prints a compact version of the Python/Rust comparison guide. `compare gemstone-js` prints the TypeScript/Python comparison from `gemstone-js vs gemstone-py`. The `--gaps` forms print remaining parity gaps with next actions and verification commands. The `--next` forms print the first recommended batch and top priority gap. The `--totals` forms print only the batch/hour totals for planning and CI. The `--batches` forms answer how much work remains, including batch counts, hour ranges, outcomes, and verification commands. Use `compare all` with the same flags to print both comparison tracks together; `compare all --batches` reports **12 batches** and roughly **86-151 hours** across the Rust and TypeScript catch-up tracks. `examples scaffold` creates standalone Cargo projects from installed templates for quickstart, browser, BridgeRoot mapping, derive mapping, generated wrappers, live discovery, profile-driven codegen, standard HTTP, worker-pool, Axum, and Actix workflows. JSON output is intended for CI, docs checks, and editor tooling. Compare reports are covered by `schemas/gemstone-rs.compare.schema.json`, so release scripts and editor panels can validate the same summary, gap, next-action, totals, and batch-plan shapes that the CLI prints.

Streams

Stream	Feature	Rust surface	Examples	Docs	gems
1	Runtime and GCI loading	<code>gemstone-gci</code> , <code>gemstone-rs::Config</code>	<code>hello</code> , <code>hello_gemstone</code> , <code>quickstart</code>	<code>docs/setup-guide.md</code> , <code>docs/performance-safety.md</code>	<code>gemstone-native</code>
2	Safe sessions and transactions	<code>gemstone-rs::Session</code> , <code>SessionWorker</code> , <code>SessionWorkerPool</code> , <code>TransactionGuard</code>	<code>quickstart</code> , <code>transactions</code> , <code>session_worker</code> , <code>session_worker_pool</code> , <code>live_smoke_cookbook</code>	<code>docs/user-manual.md</code> , <code>docs/cookbook.md</code>	<code>GemstoneSession</code> , <code>SessionTransaction</code>
3	Browser and inspection	<code>gemstone-rs::browser</code> , CLI <code>browse</code> , <code>gemstone-rs-explorer</code>	<code>browser</code> , <code>tooling/cli-browser-walkthrough.md</code>	<code>docs/user-manual.md</code> , <code>docs/explorer.md</code>	<code>gemstone-python-database</code>
4	OOP and value handling	<code>oop</code> , <code>Value</code> , export-set helpers	<code>oop_values</code>	<code>docs/user-manual.md</code> , <code>docs/performance-safety.md</code>	Managed typed
5	BridgeRoot object mapping	<code>gemstone-rs::bridge</code> , <code>gemstone-rs-macros</code>	<code>bridge_root_mapping</code> , <code>derive_mapping</code> , <code>generated_mapping_app</code>	<code>docs/object-mapping.md</code> , <code>docs/cookbook.md</code>	Small Persistent examples

Stream	Feature	Rust surface	Examples	Docs	gems
6	Typed codegen	<code>gemstone-rs::codegen</code> , CLI <code>codegen</code> and <code>profile</code>	<code>codegen_preview</code> , <code>codegen_workflow</code> , <code>codegen_discover</code> , <code>profile_codegen_workflow</code> , <code>generated_wrapper_app</code>	<code>docs/codegen.md</code> , <code>docs/profile-schema.md</code>	<code>gemstone-type-codegen</code>
7	Explorer workflow	<code>gemstone-rs-explorer</code>	<code>tooling/explorer.md</code>	<code>docs/explorer.md</code> , <code>docs/screenshots.md</code>	<code>python-database</code>
8	VS Code workbench	<code>vscode-gemstone-rs-workbench</code>	<code>tooling/vscode-workbench.md</code>	<code>docs/vscode-workbench.md</code>	<code>gemstone</code>
9	Rust web services	<code>gemstone-rs::web</code> , <code>gemstone-rs-axum</code> , <code>gemstone-rs-actix</code> , <code>SessionWorkerPool</code> , checked Axum/Actix services, and installed scaffolds	<code>session_worker</code> , <code>session_worker_pool</code> , <code>http_service</code> , <code>examples/axum-service</code> , <code>examples/actix-service</code> , <code>framework_route_smoke.py</code> , <code>examples scaffold session_worker_pool/axum_service/actix_service</code>	<code>docs/examples-guide.md</code> , <code>docs/cookbook.md</code>	FastAPI examples

Stream	Feature	Rust surface	Examples	Docs	gems
10	Release and verification	<code>scripts</code> , <code>Makefile</code> , GitHub Actions	Release verification commands	<code>docs/release-checklist.md</code>	PyPI/VSIX
11	Shared native core	<code>gemstone-gci</code> , <code>gemstone-rs</code> , future <code>gemstone-py-native</code> wrapper	Shared-core integration plan	<code>docs/shared-core-integration.md</code>	<code>gems</code>

What This Says Compared With `gemstone-py`

`gemstone-rs` is strongest where Rust is naturally valuable: direct native GemStone access, explicit OOP/value handling, typed generated wrappers, BridgeRoot mapping, and CLI/explorer tooling that can run without Python in the process.

`gemstone-py` remains stronger where Python already has mature product shape: broader web framework adapters, async examples, the database explorer, package extras, and broader release surfaces.

The projects should converge by sharing the Rust native core underneath `gemstone-py-native`, while keeping the Python and Rust APIs idiomatic for their respective users.

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